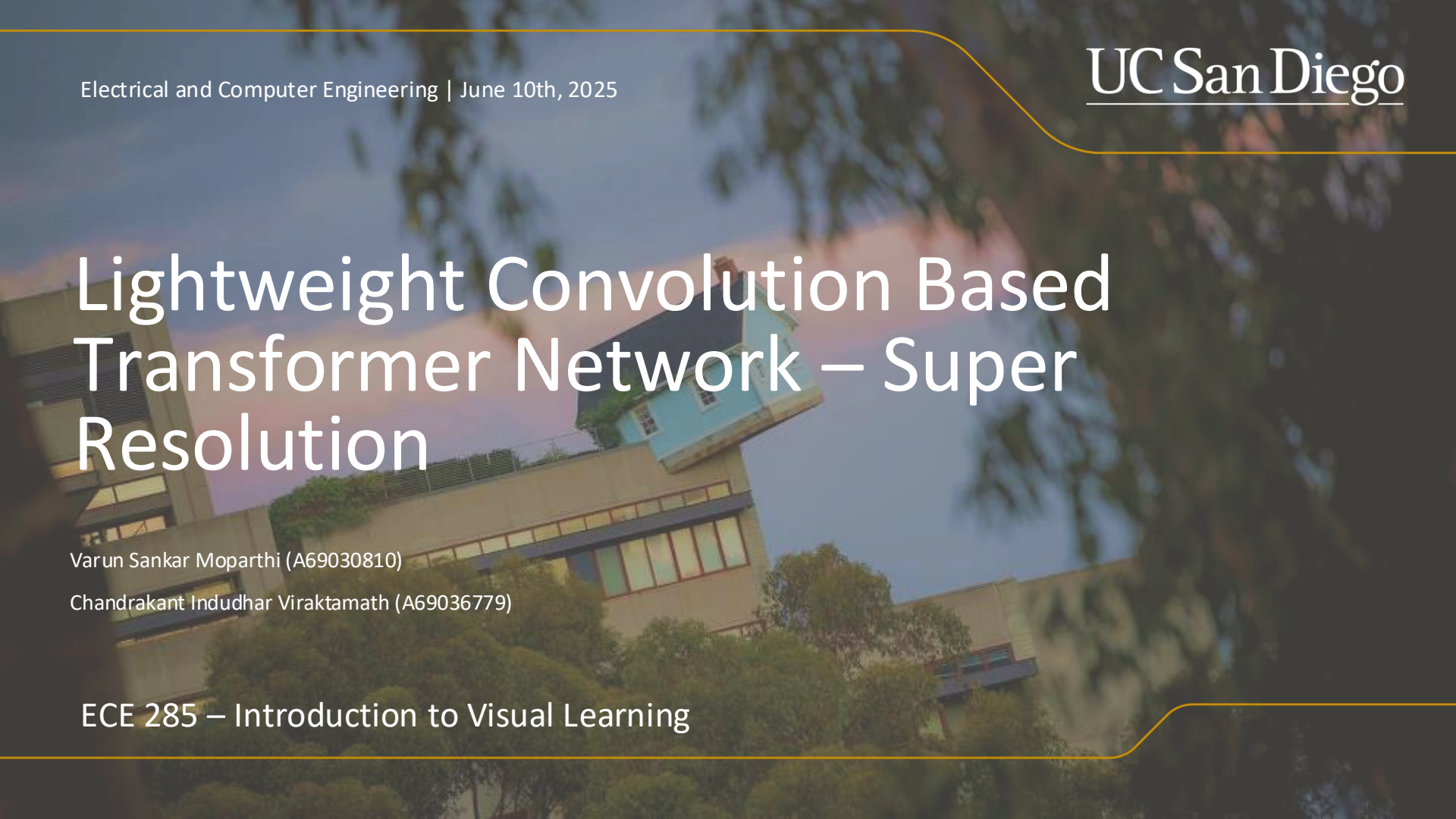


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# Lightweight Convolution Based Transformer Network – Super Resolution

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ECE 285 – Introduction to Visual Learning

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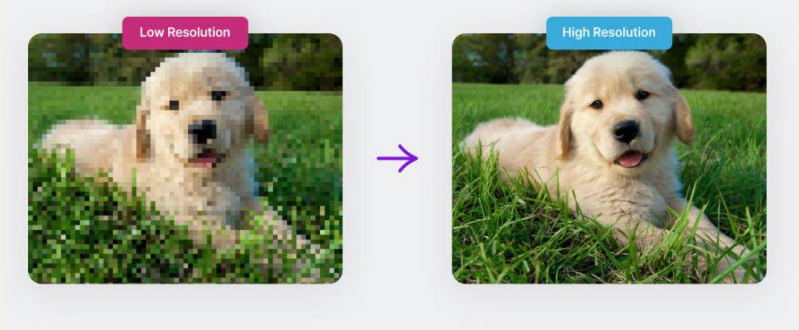
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# Super Resolution - Introduction

- **SISR** aims to reconstruct a **high-resolution (HR) image** from a single **low-resolution (LR) input** — an **ill-posed inverse problem** due to downsampling loss.




- **Key Applications:**
  - **Mobile Photography** – Enhances detail without bulky sensors
  - **Medical Imaging** – Boosts diagnostic clarity under safety limits
  - **Surveillance** – Recovers facial/object details from degraded footage
  - **Remote Sensing** – Improves geospatial resolution from satellites

# Problem Statement & Motivation

- Super-resolution models based on deep networks are typically compute- and memory-intensive, making them unsuitable for real-time deployment on edge devices like smartphones, drones, and embedded systems operating under tight resource constraints.
- **Key Challenges:**
  - Balancing model compactness with reconstruction quality
  - Capturing both fine-grained local textures and global semantic context
  - Minimizing artifacts and noise introduced during upscaling
  - Ensuring generalization across image domains and supporting real-time inference on low-resource hardware
- **Goal** - To develop lightweight yet accurate super-resolution models that can deliver high-quality visual outputs while meeting the demands of real-world, resource-constrained environments.

# Related Work

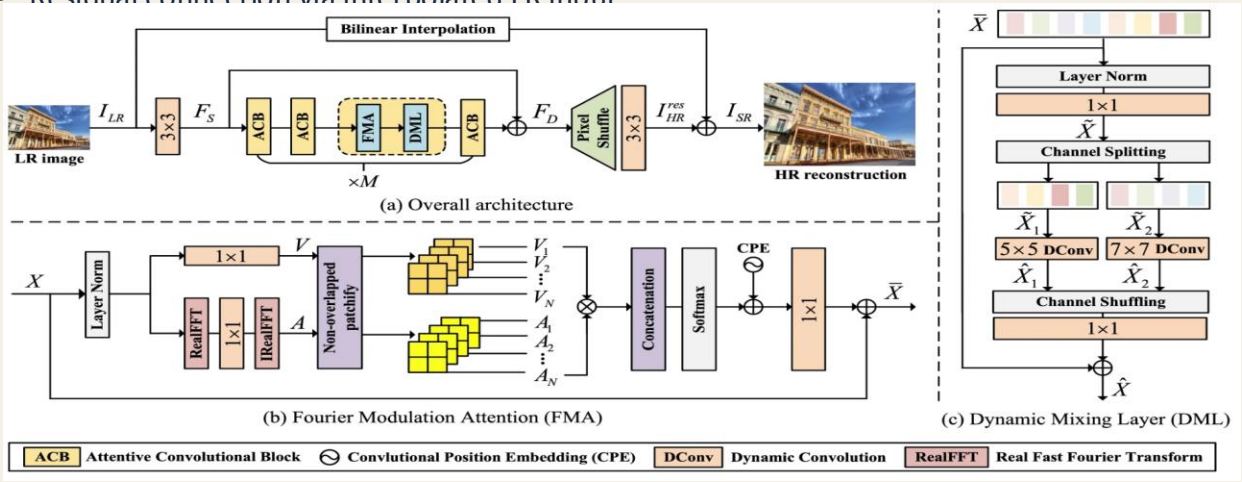
## Lightweight CNN + Transformer Approaches

- Fang et al. (HNCT) **[1]** Combines CNN with Swin Transformers and ESA blocks to balance accuracy and efficiency.
- Zou et al. (SCET) **[2]** Decouples spatial and channel modeling with channel-wise attention for efficiency.
- Garas et al. **[3]** Uses simplified Transformer-like blocks optimized for resource-constrained devices.
- Li et al. (SRConvNet) **[4]** Mimics Transformer behavior via Fourier and dynamic convolutions; fast and high-quality.

# Methodology – SRConvNet Model Overview

- **Model Architecture Highlights:**
  - Initial Conv + GELU for feature extraction.
  - Stack of ACB (Attentive Convolutional Blocks) for context-aware feature refinement.
  - Upsampling via Pixel Shuffle or Bilinear Interpolation.
  - Residual connection via interpolated LR input

- **Attention Block Details:**
  - Combines multi-head self-attention across spatial chunks.
  - Uses diverse kernel sizes (5, 7) for conv layers.

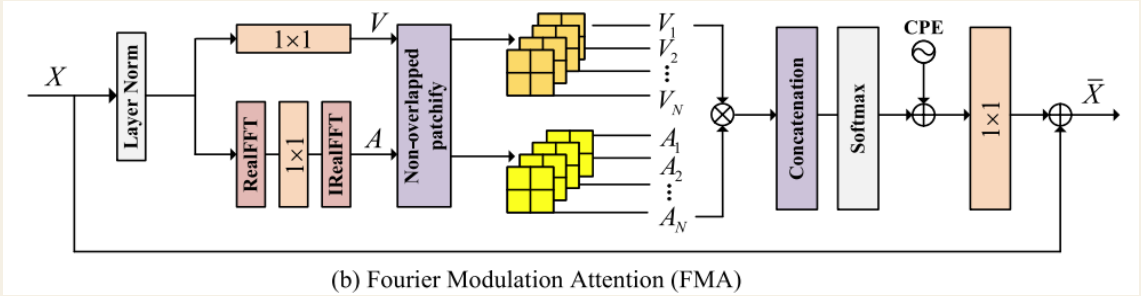





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# Methodology – SRConvNet Model Overview

- **Fourier Modulated Attention Module**
- **Purpose:** Enhances global feature modeling while being computationally efficient.
- **Design:**
  - Integrates frequency-domain modulation with attention maps.
  - Projects features into Fourier space → applies modulation → reconstructs enriched spatial representation.
- **Benefits:**
  - Captures long-range dependencies at lower cost than traditional self-attention.
  - Encourages global structure preservation during SR.



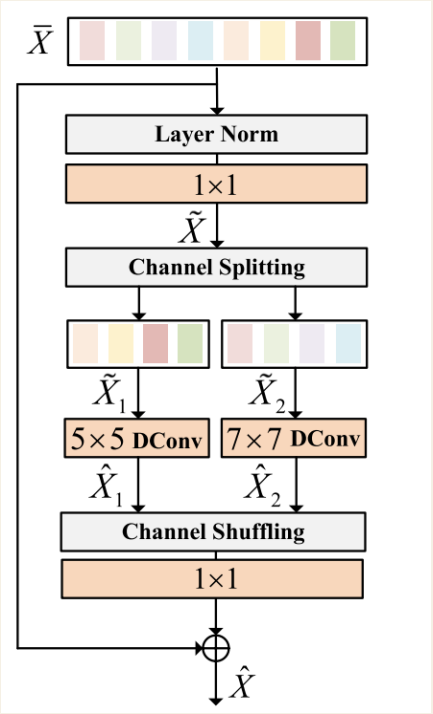
(b) Fourier Modulation Attention (FMA)



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# Methodology – SRConvNet Model Overview

- **Dynamic Multi-Scale Learning and Attentive Convolution Blocks**
- **DML Mechanism:**
  - Learns representations at multiple resolutions via dynamic reshaping and interaction.
  - Facilitates adaptive scaling and contextual aggregation.
- **ACB (Attentive Convolutional Block):**
  - Combines spatial chunking with multi-head convolutional attention.
  - Uses kernels of sizes 5, 7 over channel-chunked patches.
  - Enables fine-grained local context capture with low overhead.
- **Combined Impact:**
  - FMA+ DML ensure both local texture detail and global coherence.
  - Contributes to lightweight design without degrading visual fidelity.






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# Dataset Details

- **Training Datasets**
- **RealSR (2X, 3X, 4X scales)**
  - Real-world DSLR-captured LR-HR image pairs.
  - ~595 image pairs; multiple focal lengths and devices.
  - Resolutions vary; contains complex, real degradations.
- **DIV2K (2X, 3X, 4X scales)**
  - 800 train / 100 val images at 2K resolution (1920×1080+).
  - Downsampled using bicubic interpolation for 2×, 3×, 4× SR.
  - Clean and high-quality natural images for SR training.
- **Evaluation Dataset (2X, 3X, 4X scales)**
  - Publicly hosted on FigShare as part of a super-resolution benchmark release.
  - Classified as Set 5, Set 14, BSD 100 and Urban 100 sets.

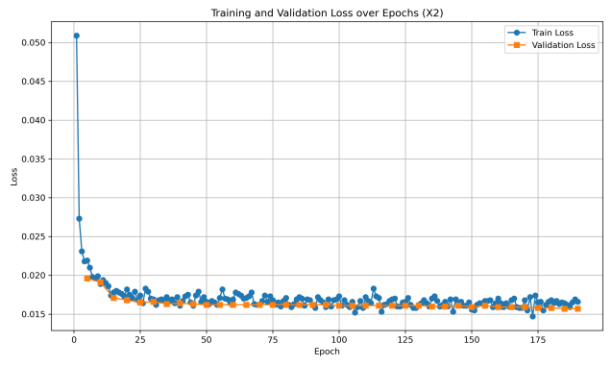
| FigShare Dataset | | | |
|---|---|---|---|
| **Dataset Type** | **#Images** | **Resolution Range** | **Description** |
| **Set5** | 5 | ~250×250 → 500×500 | Simple, clean textures; popular benchmark |
| **Set14** | 14 | ~250×250 → 500×500 | Moderate complexity, mix of scenes |
| **BSD100** | 100 | ~480×320 → 960×640 | Natural images, medium difficulty |
| **Urban100** | 100 | ~700×500 → 1400×1000 | Complex structures, rich edges; very challenging |

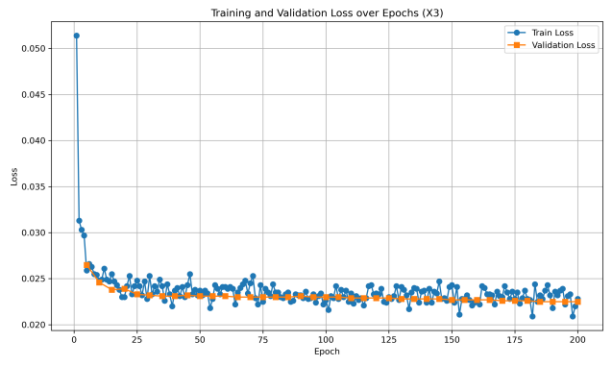
# Methodology – Training Details

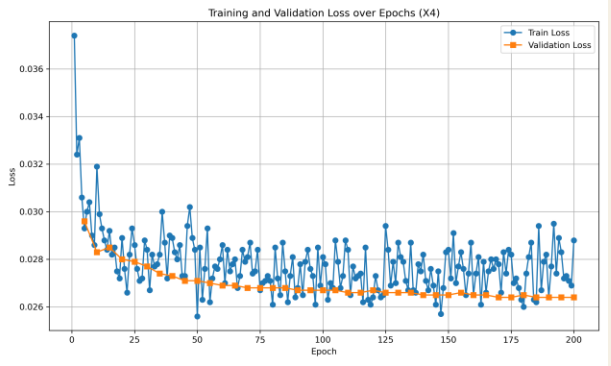
- **Training Setup:**
  - Models trained separately for 2×, 3×, and 4× upscaling tasks.
  - Mixed precision training (fp16) with PyTorch AMP.
  - **Optimizer:** Adam (LR = 1e-4, eps = 1e-8).
  - **Scheduler:** MultiStepLR (decays at 50 and 100 epochs, gamma = 0.5).
  - **Batch Size:** 2 (per GPU) with distributed training via PyTorch DDP.
  - **Loss Function:** L1 Loss on RGB images.
  - Patch-based training (patch size = 96) with data augmentation.
  - Pretrained weights of 2X models loaded for 3X and 4X model training for faster convergence

- **Validation & Setup**
  - Validation on Y-channel PSNR/SSIM every 5 epochs.
  - Results and model checkpoints saved every 5 epochs.

# Results and Analysis – Training & Validation Loss (DIV2K)



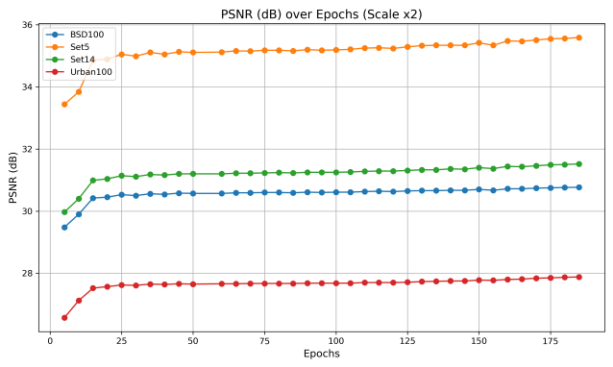



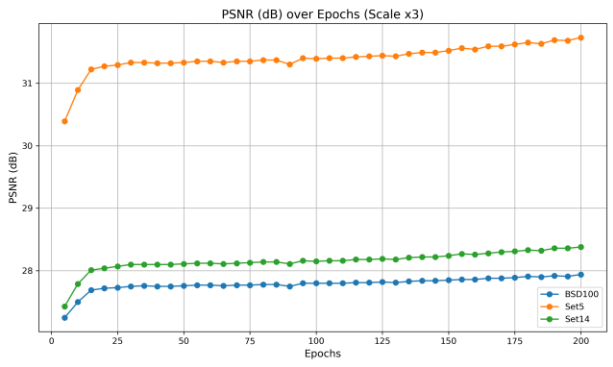

| | Loss | |
|---|---|---|
| **Model** | **Training** | **Validation** |
| **X2** | 0.0166 | 0.015 |
| **X3** | 0.0228 | 0.0225 |
| **X4** | 0.0288 | 0.0264 |

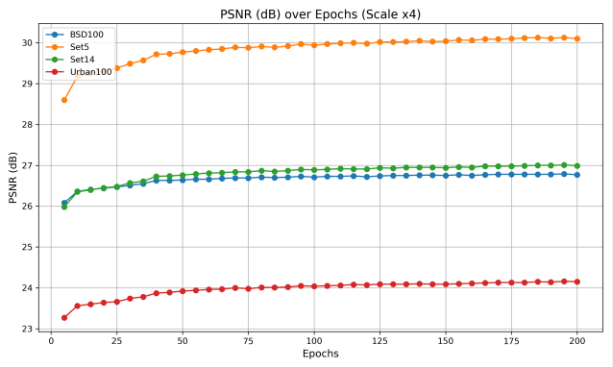


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# Results and Analysis – PSNR Data (DIV2K)



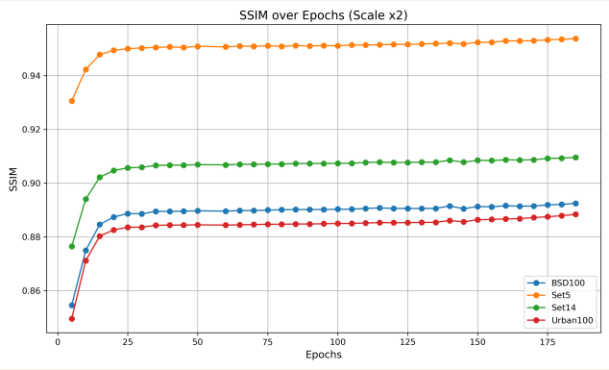



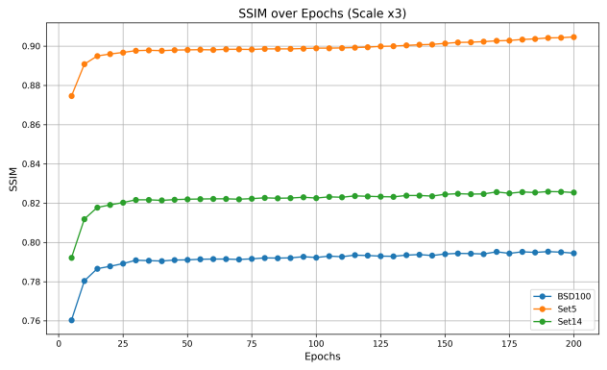

| DIV2K Dataset | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | PSNR | | | | | | | |
| | BSD100 | | Set5 | | Set14 | | Urban100 | |
| Model | Max | Final | Max | Final | Max | Final | Max | Final |
| X2 | 30.77 | 30.77 | 35.59 | 35.59 | 31.52 | 31.52 | 27.88 | 27.88 |
| X3 | 27.94 | 27.94 | 31.73 | 31.73 | 28.38 | 28.38 | N/A | N/A |
| X4 | 26.79 | 26.77 | 30.13 | 30.1 | 27.01 | 26.99 | 24.16 | 24.15 |

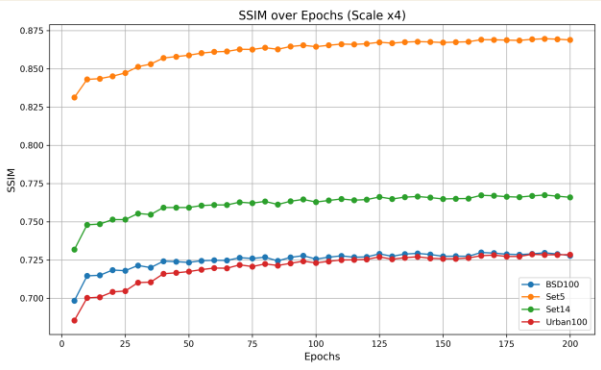


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# Results and Analysis – SSIM Data (DIV2K)

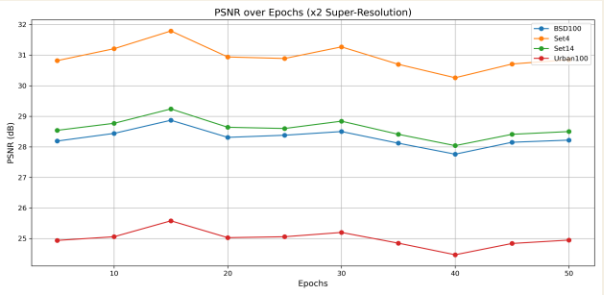


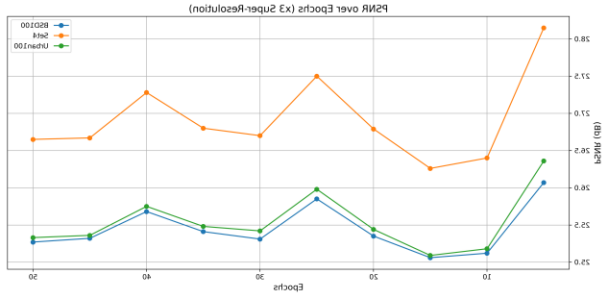



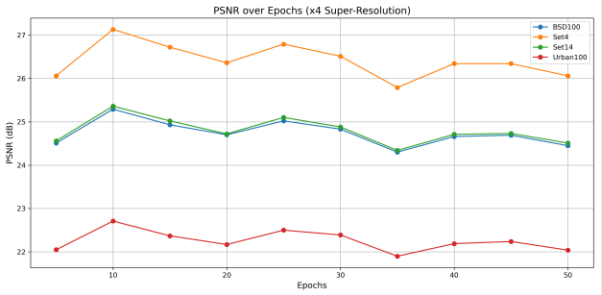




| DIV2K Dataset | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | SSIM | | | | | | | |
| | BSD100 | | Set5 | | Set14 | | Urban100 | |
| Model | Max | Final | Max | Final | Max | Final | Max | Final |
| X2 | 0.8925 | 0.8925 | 0.9538 | 0.9538 | 0.9096 | 0.9096 | 0.8884 | 0.8884 |
| X3 | 0.7945 | 0.7945 | 0.9047 | 0.9047 | 0.8254 | 0.8254 | N/A | N/A |
| X4 | 0.7289 | 0.7278 | 0.8693 | 0.8689 | 0.7667 | 0.766 | 0.7284 | 0.7285 |

# Results and Analysis – PSNR & SSIM (REALSR)

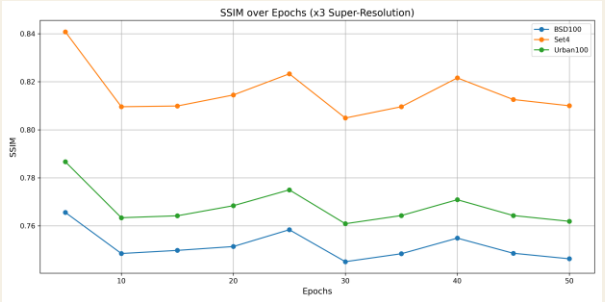












| Real SR Dataset | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Model | PSNR | | | | | SSIM | | | |
| | BSD100 | Set5 | Set14 | Urban100 | | BSD100 | Set5 | Set14 | Urban100 |
| X2 | 28.87 | 31.79 | 29.24 | 25.58 | | 0.8695 | 0.9174 | 0.8807 | 0.8494 |
| X3 | 26.07 | 28.15 | 26.36 | N/A | | 0.7656 | 0.8408 | 0.7867 | N/A |
| X4 | 25.29 | 27.13 | 25.36 | 22.71 | | 0.7141 | 0.8205 | 0.742 | 0.7 |

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# Results and Analysis – Sample Outputs

BILINEAR INTERPOLATION OUTPUTS



REALSR Trained Model Outputs

X2 X3 X4

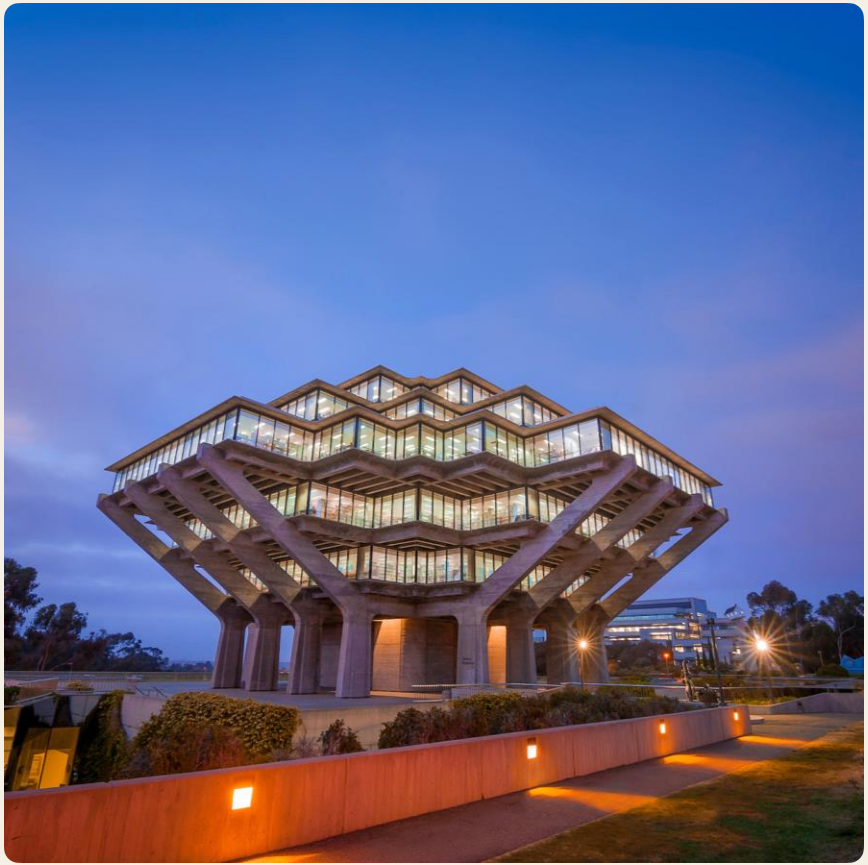


# Conclusion & Future Explorations

- **Conclusion:**
  - We implemented SRConvNet — a transformer-style ConvNet for lightweight SISR.
  - FMA (Fourier Modulated Attention) models both global and local dependencies efficiently.
  - DML (Dynamic Mixing Layer) enhances adaptability across spatial scales.
  - ACB ensures local feature learning with attention efficiency.
  - SRConvNet achieves a strong trade-off between quality and speed, well-suited for edge deployment.
- **Future Directions:**
  - Use SRConvNet to pre-process noisy, low-resolution inputs for semantic segmentation. Super-resolution can improve boundary clarity and feature integrity, leading to higher segmentation accuracy.
  - Integrate SRConvNet as a pre-processing module in object detection frameworks (e.g., YOLO, Faster R-CNN). Enhanced resolution boosts small object visibility and spatial consistency, improving detection precision in surveillance or autonomous driving scenarios.

# References


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# Thank You!